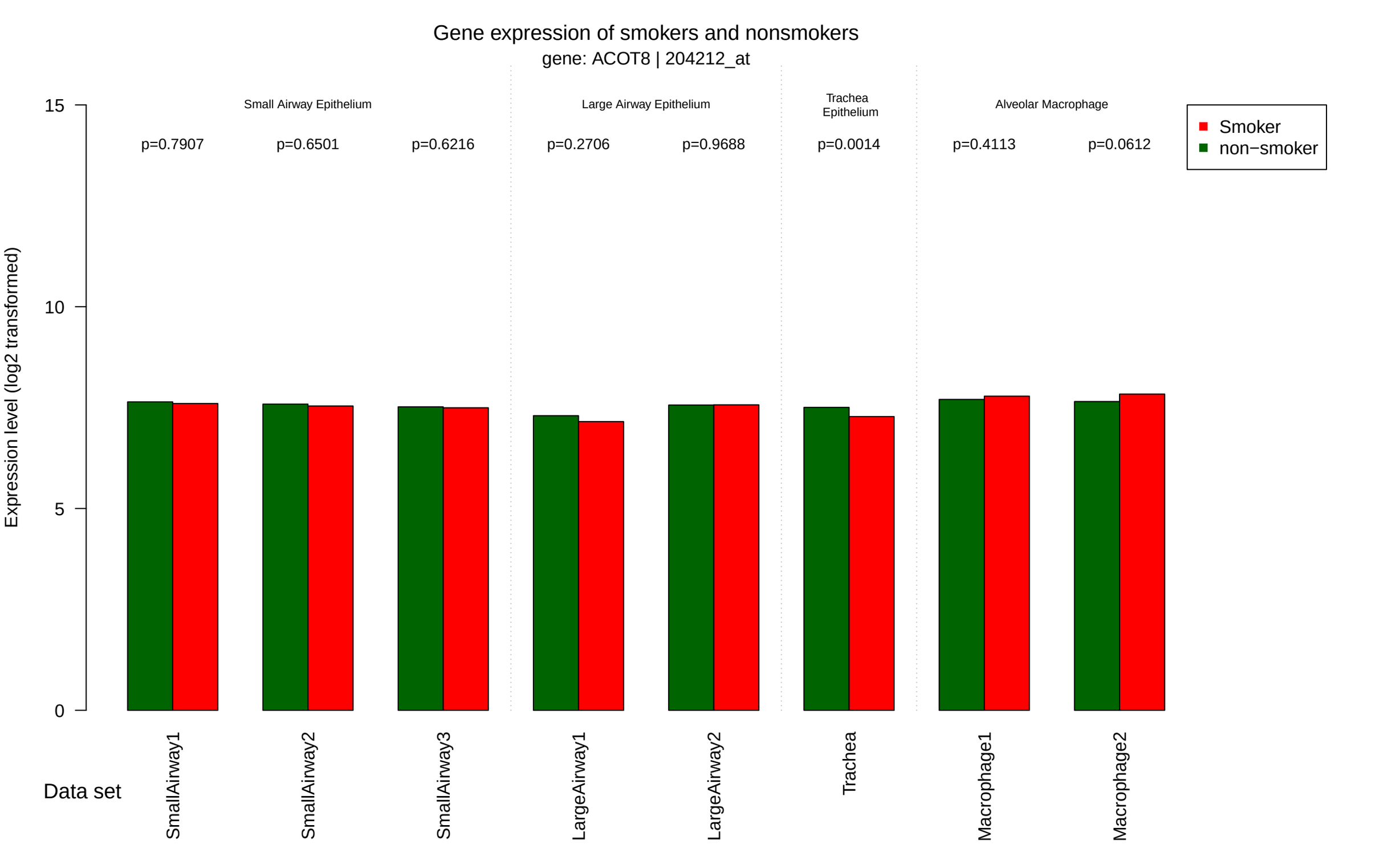
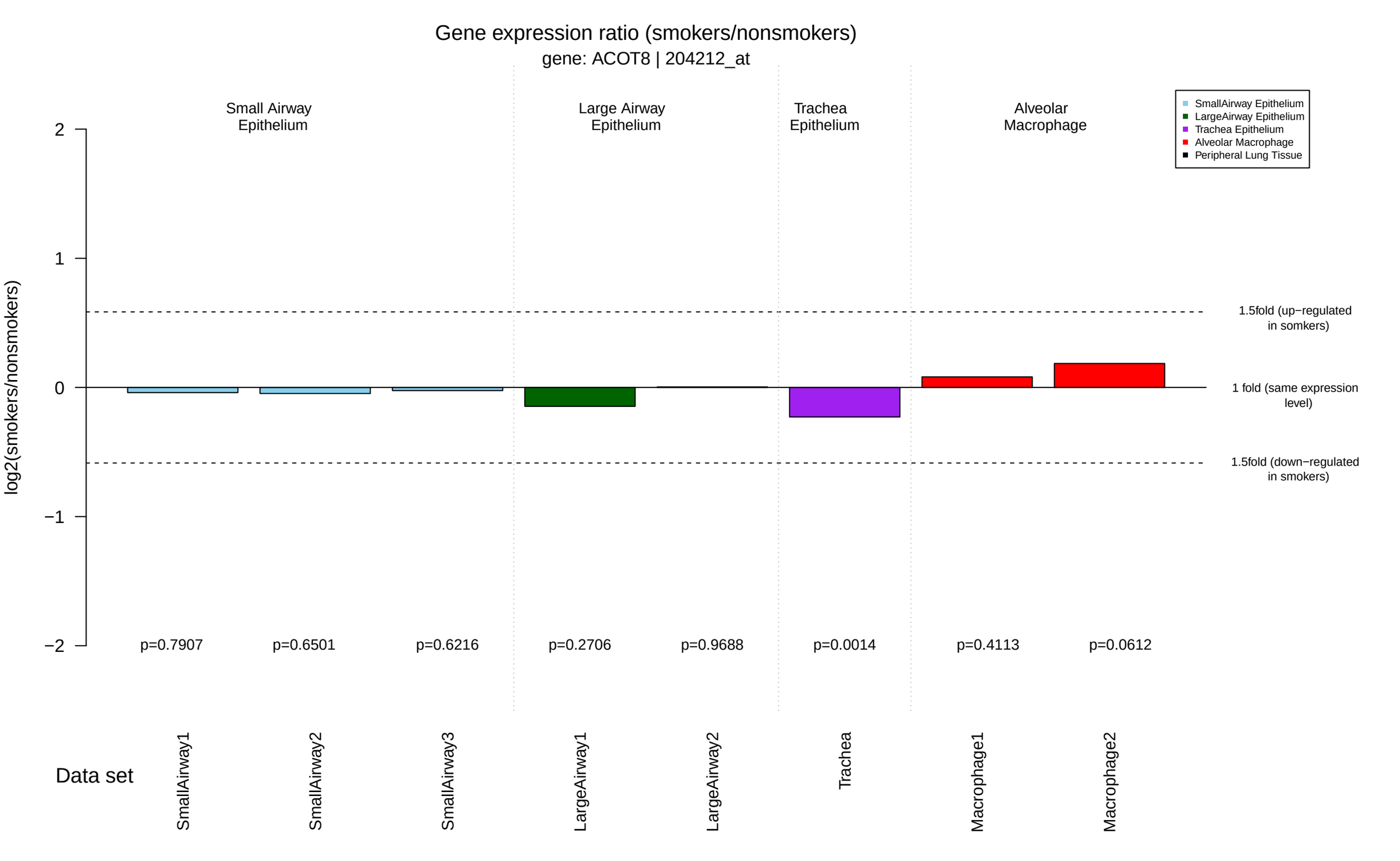
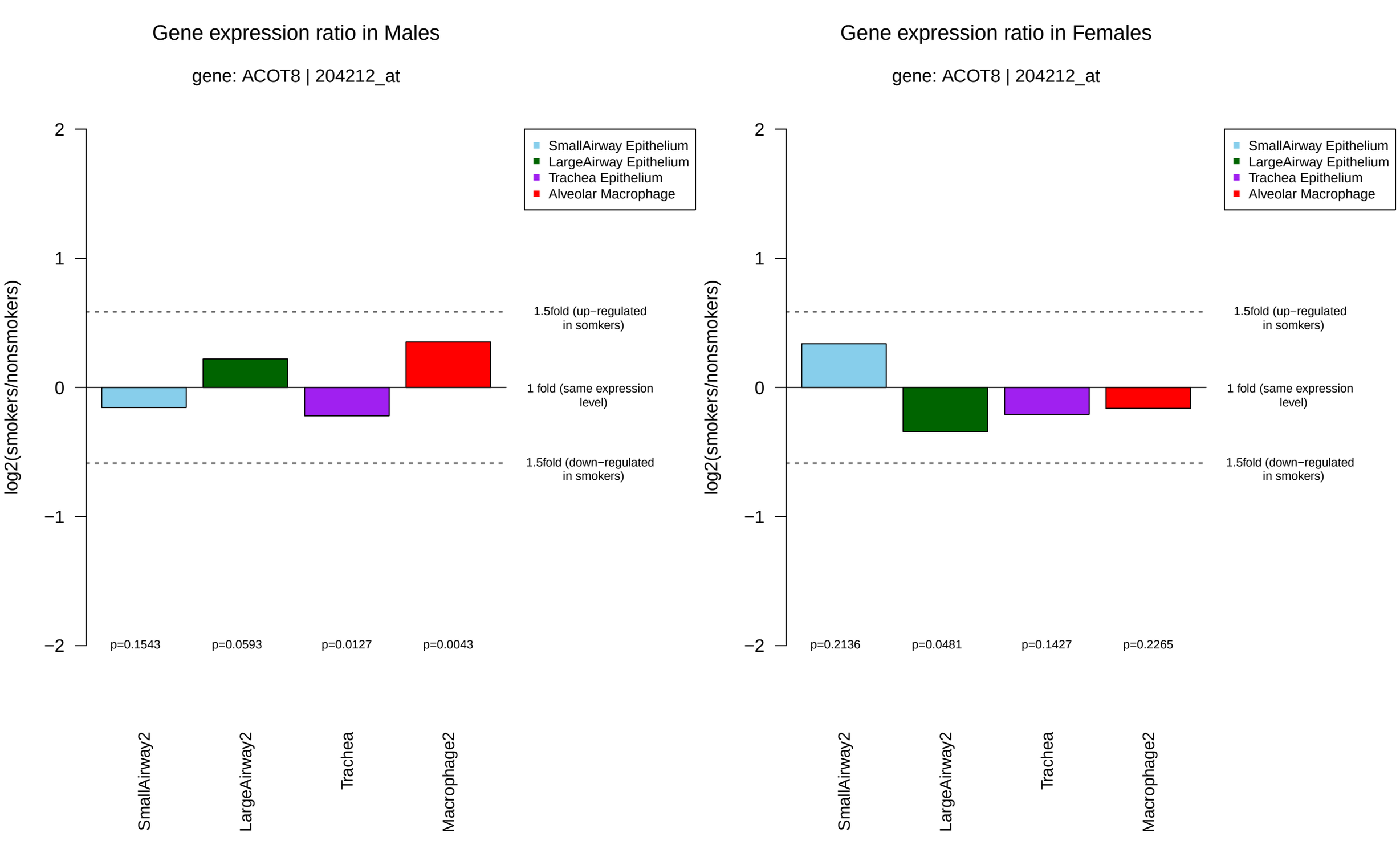
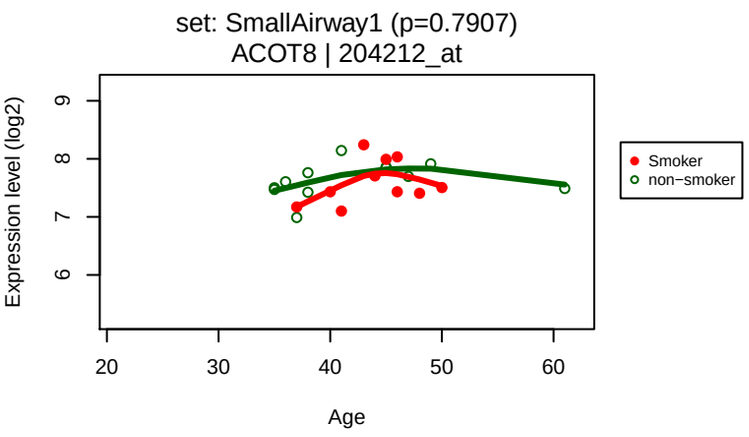
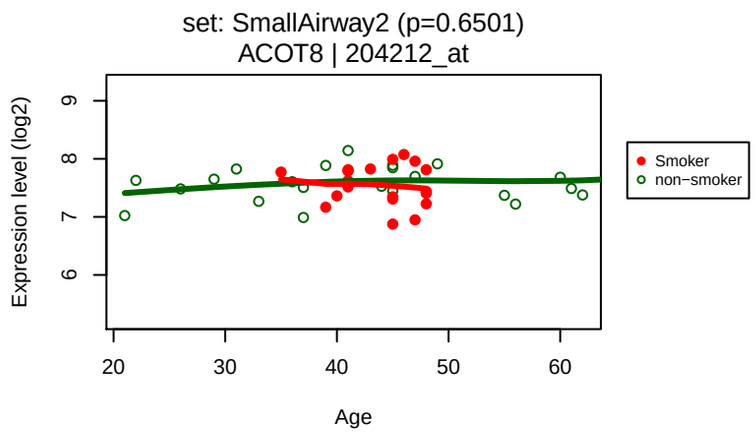
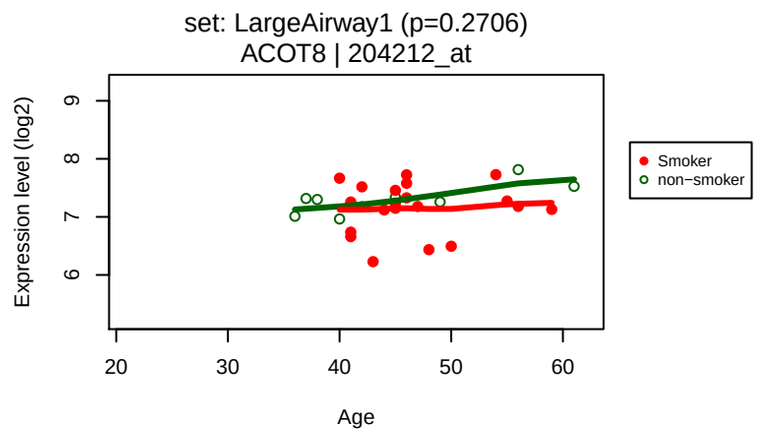

Gene expression of smokers and nonsmokers
gene: ACOT8 | 204212_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7907
p=0.6501
p=0.6216
p=0.2706
p=0.9688
p=0.0014
p=0.4113
p=0.0612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

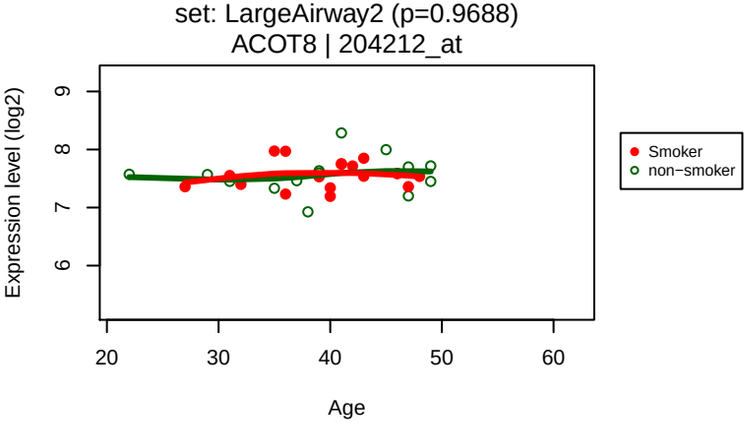

Gene expression ratio (smokers/nonsmokers)
gene: ACOT8 | 204212_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7907
p=0.6501
p=0.6216
p=0.2706
p=0.9688
p=0.0014
p=0.4113
p=0.0612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

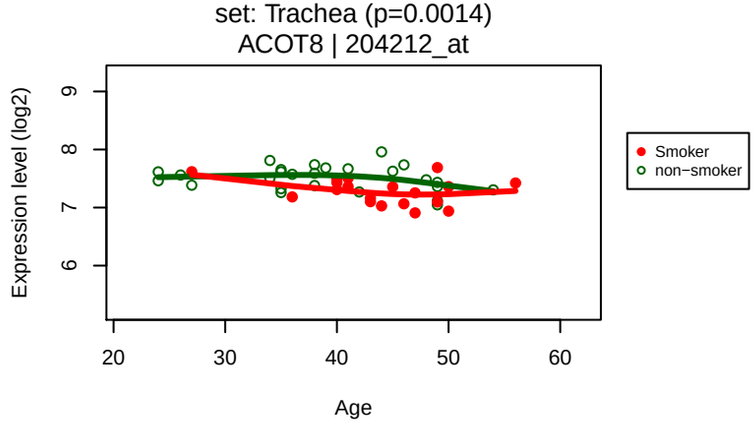

Gene expression ratio in Males
gene: ACOT8 | 204212_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1543
p=0.0593
p=0.0127
p=0.0043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOT8 | 204212_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2136
p=0.0481
p=0.1427
p=0.2265
SmallAirway2
LargeAirway2
Trachea
Macrophage2

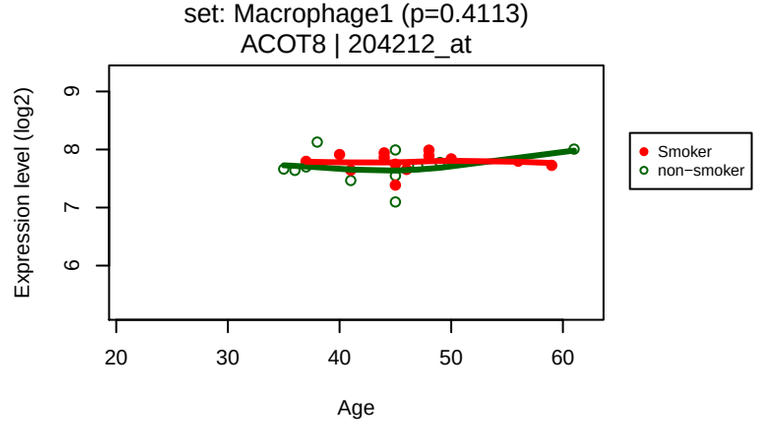

set: SmallAirway1 (p=0.7907)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker

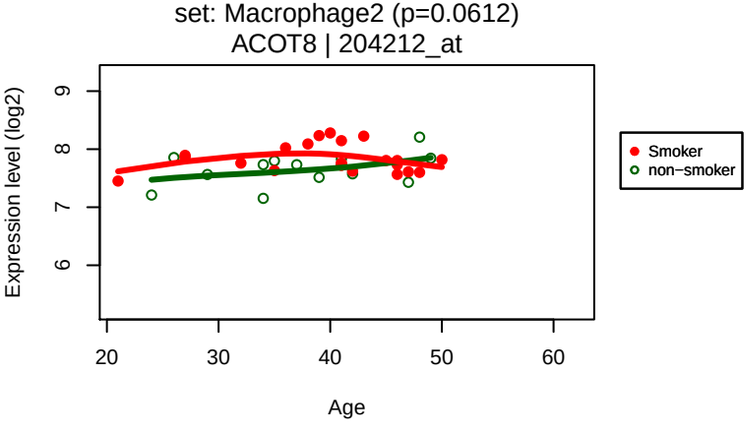

set: SmallAirway2 (p=0.6501)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker

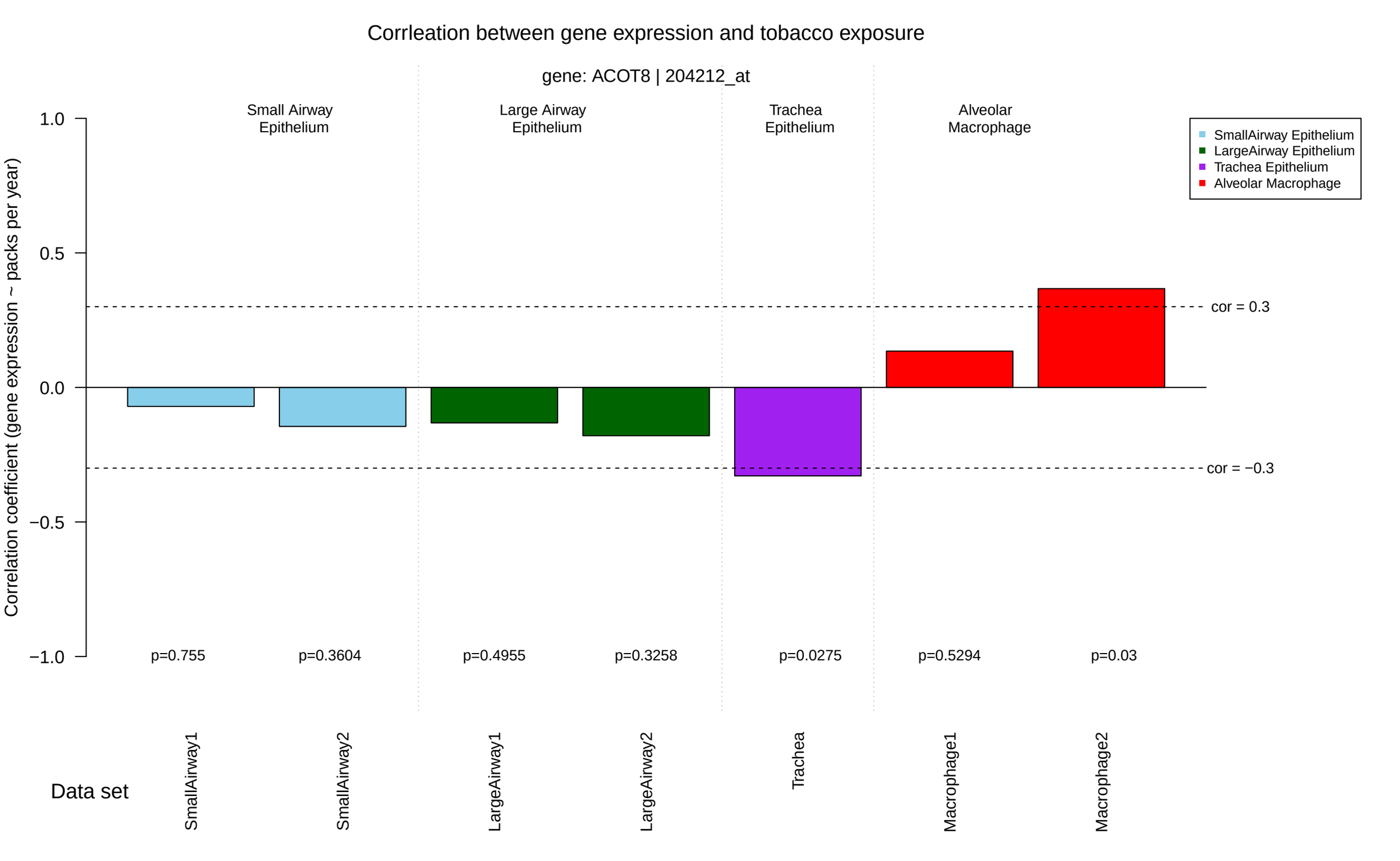

set: LargeAirway1 (p=0.2706)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9688)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4113)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0612)
ACOT8 | 204212_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOT8 | 204212_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.755
p=0.3604
p=0.4955
p=0.3258
p=0.0275
p=0.5294
p=0.03
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2